

Lung Nodule Segmentation — SegResNet 3D — wide, v1 (unified) split (legacy)

1. Task

Voxel-level segmentation of pulmonary nodules in 3D chest CT volumes, cropped to a per-series lung bounding box.

- **Input** (1, 256, 256, 256) CT crop, intensity-normalised to [0, 1], resampled from the bbox (padding = 20 vox).
- **Output** (2, 256, 256, 256) softmax logits — class 0 = background, class 1 = nodule.

2. Model

[MONAI SegResNet](#), SegResNet (residual U-Net).

Property	Value
Architecture family	SegResNet (residual U-Net)
Spatial dims	3
Input channels	1
Output channels	2 (softmax)
Initial feature width	32 (wide)
Encoder blocks per level	[1, 2, 2, 4, 4]
Decoder blocks per level	[1, 1, 1, 1]
Dropout	0.1
Trainable parameters	82,637,282

3. Data

Trained on the **legacy unified split** (patient-grouped, dataset-stratified, **full corpus**: NLST + NSCLC + LIDC-IDRI).

Split sizes: **1 609 train / 345 val / 351 test (held out)**.

Note on split. This model uses the *first-generation* unified split. Every other two-stage nodule and joint model in the demo uses the newer unified_v2 split, and the two splits have different patient assignments. **This model's test-set numbers are NOT directly comparable to the numbers of any other model in the demo** — cross-split ranking is not valid. Kept as an architectural reference point; see segresnet_wide_v2 for the wide-SegResNet result on the current benchmark.

Class imbalance: nodule voxels are on the order of $\sim 10^{-5}$ of the total, which shapes how each metric should be read (§7).

4. Training

Setting	Value
Loss	Focal Tversky + weighted CE ($\alpha=0.3$, $\beta=0.7$, $\gamma=2.0$, $\lambda_{CE}=0.1-0.3$, <code>ce_nodule_weight = 100</code>)
Optimizer	Adam
Learning rate	1×10^{-5}
Weight decay	1×10^{-5}
LR schedule	Cosine annealing, $T_{max} = 1000$, $\eta_{min} = 1 \times 10^{-6}$
Batch size	2
Epochs	1000
Random seed	42
Mixed precision	bf16 (autocast, no GradScaler)
Augmentation	3D flips, 90° rotations, elastic rotation, zoom, intensity scale/shift, Gaussian noise/blur, contrast
Hardware	1 × NVIDIA H100 94 GB
Wall-clock	≈ 12 days

Best checkpoint (evaluated below): **epoch 544 / 1000**.

5. Evaluation protocol

Held-out test split (351 series). The training loop used the `val` split of `unified` for early stopping and best-Dice checkpoint selection; the numbers below come from the completely untouched `test` split (never seen during training or model selection). Predictions taken as `argmax` over the 2-channel softmax output; foreground = class 1. All metrics are computed at the voxel level and micro-averaged over the whole test split (5,888,802,816 voxels total).

6. Results

Ticket-required metrics

Metric	Value
mean IoU	0.7402
Accuracy	0.9996

Metric	Value
Precision	0.6465
Recall	0.6525

Supplementary

Metric	Value
IoU (foreground / nodule class)	0.4809
IoU (background class)	0.9996
Dice / F1 (micro)	0.6495

Per-case Dice distribution (351 val cases)

Statistic	Value
Mean	0.5634
Std	0.2576
Min	0.0000
p05	0.0033
p25	0.3916
Median	0.6308
p75	0.7833
p95	0.8652
Max	0.9330

7. Notes on interpretation

Nodule segmentation is severely class-imbalanced ($\sim 10^{-5}$ of voxels are nodule), which distorts the standard metric set:

- **Accuracy** is trivially near 1.0 for any reasonable model — the model gets $\approx 99.9556\%$ of voxels right by predicting "background" almost everywhere. Uninformative on its own here.
- **mIoU** averages the foreground and background IoU. `IoU_background` is essentially 1.0, so mIoU is roughly $0.5 + 0.5 \cdot \text{IoU_nodule}$. The useful signal is in `IoU_nodule` (0.4809) and Dice / F1 (0.6495).
- **Precision / Recall** are the standard per-voxel figures — no imbalance caveat needed. Recall 0.6525 means the model correctly labels $\sim 65\%$ of nodule voxels; Precision 0.6465 means $\sim 65\%$ of predicted-nodule voxels are true nodule.

Per-case Dice mean (0.5634) is lower than the micro Dice (0.6495) because large nodules dominate the micro average; per-case Dice weights each patient equally regardless of nodule size. Clinically the per-case distribution is the more useful summary.

8. How to load & run inference

Pre-trained weights are hosted at [szabopeter/nodule-segresnet-3d-wide-v1](https://huggingface.co/szabopeter/nodule-segresnet-3d-wide-v1). Download and load directly:

```
huggingface-cli download szabopeter/nodule-segresnet-3d-wide-v1 --local-dir ./ckpt
```

```
import yaml, torch
from monai.networks.nets import SegResNet

cfg = yaml.safe_load(open("config.yaml"))["model"]
model = SegResNet(
    spatial_dims = cfg["spatial_dims"],
    in_channels   = cfg["in_channels"],
    out_channels  = cfg["out_channels"],          # 2
    init_filters  = cfg["init_filters"],
    blocks_down  = tuple(cfg["blocks_down"]),
    blocks_up    = tuple(cfg["blocks_up"]),
    dropout_prob = cfg["dropout_prob"],
)
state = torch.load("model.pth", map_location="cpu", weights_only=True)
model.load_state_dict(state)
model.eval()

with torch.no_grad():
    # Input: (B, 1, 256, 256, 256), lung-bbox-cropped CT resampled to 256³.
    x = torch.randn(1, 1, 256, 256, 256)
    logits = model(x)                    # (B, 2, D, H, W)
    pred_class = logits.argmax(dim=1)    # (B, D, H, W) in {0, 1}
    nodule_mask = (pred_class == 1).to(torch.uint8)
```

This model expects **lung-bbox-cropped** input. A separate 2D ROI model is needed to produce that bbox — see the accompanying ROI checkpoints ([szabopeter/roi-segresnet-2d](https://huggingface.co/szabopeter/roi-segresnet-2d) or [szabopeter/roi-swinunetr-2d](https://huggingface.co/szabopeter/roi-swinunetr-2d)), or use the demo's `inference.py` which chains them for you.

9. Reproducibility

- Config `configs/segresnet_wide_v1.yaml`
- Checkpoint `checkpoints/segresnet_wide_v1/best_model.pth`
- Metrics JSON `checkpoints/segresnet_wide_v1/eval_metrics_nodule_test.json`
- Command `python eval_metrics_nodule.py --config configs/segresnet_wide_v1.yaml --ckpt checkpoints/segresnet_wide_v1/best_model.pth --split test`

Companion two-stage nodule models on HuggingFace:

- [szabopeter/nodule-segresnet-3d-wide-v2](#) — wide SegResNet on the current v2 benchmark (recommended)
- [szabopeter/nodule-segresnet-3d-small](#) — small SegResNet on v2
- [szabopeter/nodule-dynunet-3d](#) — DynUNet (3D U-Net) on v2